AITS
ALL INDIA TEST SERIES

Dropper JEE Main

Additional Test

DURATION : 180 Minutes | **DATE : 16/01/2025** | **M.MARKS : 300**

ANSWER KEY

PHYSICS

1. (1)
2. (3)
3. (4)
4. (2)
5. (2)
6. (3)
7. (2)
8. (2)
9. (3)
10. (3)
11. (3)
12. (2)
13. (2)
14. (3)
15. (3)
16. (4)
17. (3)
18. (3)
19. (4)
20. (2)
21. (4)
22. (15)
23. (40)
24. (0)
25. (64)

CHEMISTRY

26. (2)
27. (2)
28. (3)
29. (3)
30. (2)
31. (2)
32. (4)
33. (1)
34. (3)
35. (4)
36. (1)
37. (2)
38. (1)
39. (2)
40. (2)
41. (2)
42. (4)
43. (4)
44. (3)
45. (1)
46. (2)
47. (63)
48. (1)
49. (3)
50. (8)

MATHEMATICS

51. (3)
52. (2)
53. (4)
54. (2)
55. (2)
56. (2)
57. (2)
58. (2)
59. (2)
60. (1)
61. (3)
62. (3)
63. (1)
64. (3)
65. (4)
66. (2)
67. (3)
68. (1)
69. (4)
70. (1)
71. (15)
72. (7)
73. (5)
74. (10)
75. (0)

SECTION-I (PHYSICS)

1. **(1)**

$2\pi\sqrt{\frac{M}{K}}$ At the mean position of plate *A* spring will be elongated by $\frac{Q^2}{2A\varepsilon_0 K}$

2. **(3)**

$\frac{X}{R_0}=\frac{\ell}{100-\ell}$

Since null point remains unchanged

$\frac{X}{R'}=\frac{40}{60}\Rightarrow R'=6\Omega$

And $6=\frac{78R_t}{R_t+78}\Rightarrow R_t=6.5\Omega$

$\alpha=\frac{R_t-R_0}{R_0 t}=8.3\times10^{-4}\ \text{K}^{-1}$

3. **(4)**

$N=mg-T\sin\theta$

$T\cos\theta-\mu N=ma$

4. **(2)**

Let initial pressure of the three samples be P_A, P_B and P_C.

$P_A(V)^{3/2}=(2V)^{3/2}P$

$P_B=P$

$P_CV=P(2V)$

$P_A:P_B:P_C=(2)^{3/2}:1:2$

5. **(2)**

The effective focal length of the given combination is $\frac{1}{f_{\text{eff}}}=\frac{2}{f_L}+\frac{1}{f_M}\Rightarrow f_{\text{eff}}=\frac{15}{2}$ cm

The effective combination behaves as a mirror therefore we can calculate the location of final image using mirror formula.

$\frac{1}{f}=\frac{1}{u}+\frac{1}{v}$

6. **(3)**

Electric field lines inside the sphere A and B is not going to change

7. **(2)**

$JL=\frac{2ML^2}{3}\omega\Rightarrow\omega=\frac{3J}{2ML}$

8. **(2)**

$I_{Xx}=\frac{MR^2}{4}+2\left[\frac{MR^2}{4}+MR^2\right]$

$=\frac{MR^2}{4}+\frac{5}{2}MR^2=\frac{11}{4}MR^2$

9. **(3)**

$Ig\times Rg=(I-Ig)R$, putting values we get

$R=0.1\ \Omega$

10. **(3)**

$\frac{\partial y}{\partial t}=\frac{10}{\pi}\times2000\pi\cos\left(2000\pi t\cdot\frac{\pi x}{17}\right)$

= 200 m/s

11. **(3)**

$dQ=dU+dW\qquad\Rightarrow\mu CdT=\mu C_V dT+PdV$

$\Rightarrow C=C_v+\frac{PdV}{\mu dT}=C_v+\frac{PdV}{\frac{PV}{RT}dT}=C_v+\frac{RT}{V}\frac{dV}{dT}$

$=\frac{3R}{2}+\frac{RT}{V_2}\times\frac{1}{2K}=\frac{3}{2}R+\frac{R}{2}=2R$

12. **(2)**

For COM, $\frac{3m}{4}\times x=\frac{3}{4}\times\frac{R}{2}\Rightarrow x=\frac{R}{6}$

$I_{1\ (\text{complete disc})}=\frac{mR^2}{2}+mx^2$

$=\frac{mR^2}{2}+\frac{mR^2}{36}=mR^2\left[\frac{19}{36}\right]$

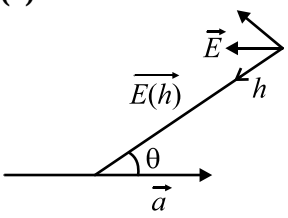
$I_{2\ (\text{colour disc})}=\frac{m}{4}\times\left(\frac{R}{2}\right)^2\times\frac{1}{2}\times\frac{m}{4}\times\left(\frac{R}{6}\times\frac{I}{2}\right)^2$

$=\frac{m}{4}\left(\frac{R}{2}\right)^2\times\frac{1}{2}+\frac{M}{4}\left(\frac{R}{6}+\frac{R}{2}\right)^2$

$=\frac{mR^2}{4}\times\frac{41}{72}=\frac{41}{288}mR$

$I=I_1-I_2=\frac{111}{288}mR^2$

13. **(2)**



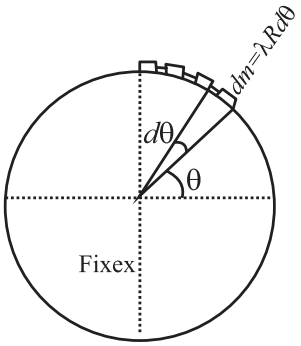
$V=ar\cos\theta$

$E_{(R)}=-\left(\frac{\partial V}{\partial r}\right)_0=\frac{\partial}{\partial r}(ar\cos\theta)=-a\cos\theta$

$E_{(0)}=\frac{\partial V}{r(\partial\theta)_r}=+\frac{1}{r}\times ar\sin\theta=a\sin\theta$

$\vec{E}=-\vec{a}$

14. (3)



$dF_T = dmg.\cos\theta$

$\lambda.R\cos\theta d\theta g$

$$F_T = \int dF_T = \lambda Rg \int_{45°}^{90°} \cos\theta d\theta$$

$$a_T = \frac{F_T}{m} = \frac{\lambda Rg}{\lambda \times 2\pi R/8}\left[1-\frac{1}{\sqrt{2}}\right] = \frac{4g}{\pi}\left[\frac{\sqrt{2}-1}{\sqrt{2}}\right]$$

15. (3)

$$k_{(x)} = k + \left(\frac{4k-k}{d}\right).x = k + \frac{3k}{d}x$$

$$\Rightarrow dC = \frac{\varepsilon_0 A k_{(x)}}{dx}$$

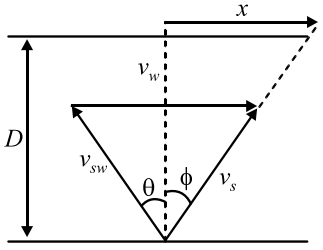
$$\Rightarrow \frac{1}{C} = \int \frac{1}{dC} = \int \frac{dx}{\varepsilon_0 A k_{(x)}} = \frac{1}{\varepsilon_0 A}\int_{x=0}^{d} \frac{dx}{k+\frac{3k}{d}x}$$

$$= \frac{1}{\varepsilon_0 A}.\frac{\log_e\left(k+\frac{3k}{d}x\right)_0^d}{\frac{3k}{d}}$$

$$\Rightarrow \frac{1}{C} = \frac{d}{3\varepsilon_0 kA}.\log_e 4 = \frac{2d\log_e 2}{3\varepsilon_0 kA}$$

$$\Rightarrow C = \frac{3\varepsilon_0 kA}{2d\log_e 2}$$

16. (4)



$v_{sw}\cdot\cos\theta = v_s\cos\phi$

$v_w - v_{sw}\cdot\sin\theta = v_s\sin\phi$

$$\tan\phi = \frac{v_w - v_{sw}\sin\theta}{v_{sw}\cos\theta}$$

$$= \frac{v_w}{v_{sw}}\sec\theta - \tan\theta$$

$$x = D\tan\phi = D\left[\frac{v_w}{v_{sw}}\cdot\sec\theta - \tan\theta\right]$$

$$\Rightarrow \frac{dx}{d\phi} = D\left[\frac{v_w}{v_{sw}}\sec\theta\tan\theta - \sec^2\theta\right] = 0$$

$$\Rightarrow \frac{v_w}{v_{sw}}\cdot\tan\theta - \sec\theta = 0$$

$$\Rightarrow \sin\theta = \frac{v_{sw}}{v_w} = \frac{2}{3}$$

So the angle with stream direction is equal to 90° + $\sin^{-1}2/3$

17. (3)

$$v_{\text{rell}|} = v + v\cos 60^\circ = \frac{3v}{2}$$

$$v_{\text{rel}\perp} = v\sin 60^\circ = \frac{v\sqrt{3}}{2}$$

$$a_N = \frac{v_{\text{rel}\perp}^2}{r} = \frac{v^2\times 3}{4\times \ell/2} = \frac{3v^2}{2l}$$

18. (3)

Let the plank shifts to left by distance x

$$M(L-x) = 2Mx \Rightarrow x = \frac{L}{3}\lim_{x\to\infty}$$

So distance from $O = \frac{L}{3}+\frac{L}{2} = \frac{5L}{6}$

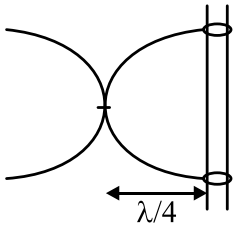
19. (4)

$$Z = \sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}$$

$\Rightarrow$ As we introduce dielectric, capacitance increases reducing the impedance of the circuit.

$\Rightarrow$ I_{rms} increases

20. (2)



$$\lambda = \frac{36\text{ m/s}}{30} = 1.2\text{ m}$$

$\Rightarrow$ 0.3 m

21. (4)

As voltage across C is 30 V, voltage across 3 μF must be 60 V which implies both the capacitors have same charge i.e., the bridge consisted of capacitors is balanced

$\Rightarrow \frac{3}{6} = \frac{2}{X}$ i.e., $X = 4\mu F$

22. (15)

$$e_{yz} = e_{xyz} - e_{xy} = \frac{1}{2}B\omega(2R)^2 - \frac{1}{2}B\omega(R\sqrt{2})^2$$

$$= \frac{1}{2}B\omega\left(4R^2 - 2R^2\right) = B\omega R^2$$

$$\Rightarrow e = 3\times 5\left(\frac{10}{100}\right)^2 = 15\times 10^{-2} \text{ volt}$$

23. (40)

$$I_{max} = Q\cdot\omega = C\cdot V\cdot\frac{1}{\sqrt{LC}} = v_0\sqrt{\frac{C}{L}}$$

24. (0)

$$T_A = \frac{2P_0V_0}{R}; T_B = \frac{2P_0V_0}{R}$$

$\therefore T_A - T_B = 0$

$\therefore \Delta U = 0$

25. (64)

$$|E| = \frac{Mdl_1}{dt} = M\alpha$$

Then $I_2 = \frac{E}{R}\left[1 - e^{-\frac{R}{L}t}\right] = \frac{M\alpha}{R}\left[1 - e^{-\frac{R}{L}t}\right]$

$$= \frac{M\alpha}{R}\left[1 - e^{-1}\right] = \frac{M\alpha}{R}\left[1 - \frac{1}{e}\right] = \frac{M\alpha}{R}\left[\frac{e-1}{e}\right]$$

$$= \frac{0.64M\alpha}{R}$$

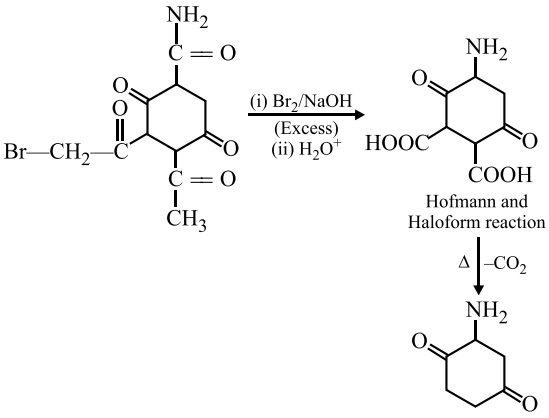
SECTION-II (CHEMISTRY)

26. (2)

$KMnO_4 + FeSO_4 + H^+ \rightarrow Fe^{3+} + Mn^{2+}$

Yellow colour due to Fe^{3+}.

27. (2)



28. (3)

The rate of the third bromination is greater than the rate of the second bromination and the rate of the second bromination is greater than the rate of the first bromination.

29. (3)

Since CN^- and NO^+ are iso eletronic, they have bond order equal to 3 as per MOT.

30. (2)

For Balmer series, the electron de-excites from nth orbit to 2nd orbit. The wave number of emission spectrum of H-atom in Balmer series is given by

$$\bar{v} = R\left[\frac{1}{(2)^2} - \frac{1}{(n)^2}\right] = \frac{R(n+2)(n-2)}{4n^2}$$

31. (2)

$[NH_3] = 0.3$ M $\qquad [NH_4^+] = 0.2$ M

$$pOH = pK_b + \log\frac{[Salt]}{base} = 4.74 + \log\frac{2}{3}$$

= 4.74 + 0.30 – 0.48 = 4.56

pH = 14 – 4.56 = 9.44

32. (4)

$$-COOH + HCO_3^- \rightarrow -COO^- + CO_2\uparrow + H_2O$$

$$-SO_3H + HCO_3^- \rightarrow SO_3^{2-} + CO_2\uparrow + H_2O$$

$$-CH_2OH + HCO_3^- \rightarrow \text{ No reaction}$$

$$-COOH + Na \rightarrow -COO^-\,Na^+ + \frac{1}{2}H_2$$

$$-SO_3H + Na \rightarrow -SO_3^-\,Na^+ + \frac{1}{2}H_2$$

$$-CH_2OH + Na \rightarrow -CH_2O^-\,Na^+ + \frac{1}{2}H_2$$

33. (1)

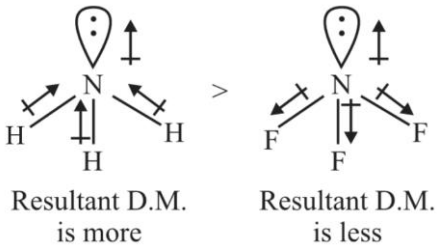
Both are functional isomers not metamers

34. (3)

Effect of temperature on equilibrium constant Based on Le Chatelier's principle

35. (4)

Both NF_3 and NH_3 have identical shape and a lone pair of electron on N-atom

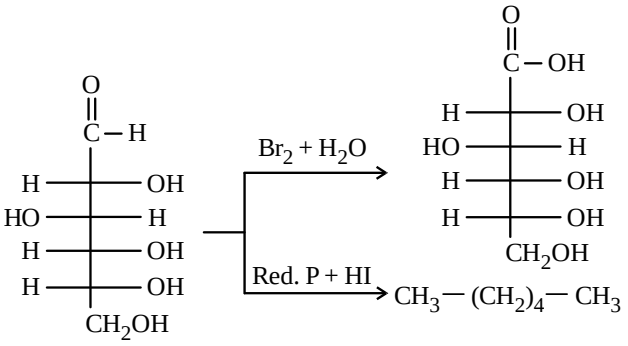


Resultant D.M. is more | Resultant D.M. is less

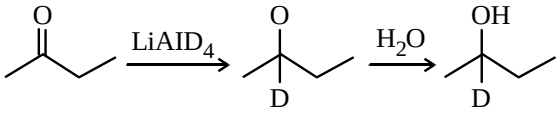
BF_3 has zero D.M. being symmetrical molecule

Hence correct order is $NH_3 > NF_3 > BF_3$

36. (1)



37. (2)



38. (1)

$\Delta H = \Delta U + P_2V_2 - P_1V_1$

$\Delta U = q + w$

$q = 2740 kJ$

$w = -\text{Area of plot} = -\left[\frac{1}{2} \times 2 \times 9 + 1 \times 2\right] \times 10^5 \text{ Pa m}^3$

$= -1100$ kJ

$\Delta U = 2740 - 1100 = 1640 kJ$

$\Delta H = 1640 + [1 \times 4 - 10 \times 2] \times 100 = 40 kJ$

39. (2)

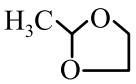
PCl_4^+ – sp^3 hybridization $\rightarrow$ Tetrahedral

PCl_4^- – sp^3d dhybridization with 1 $\rightarrow$ lp see saw

$AsCl_3$ – sp^3 hybridization with 1 lp $\rightarrow$ Pyramidal

40. (2)

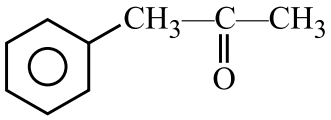
–CHO, and > C = 0 are reduced by $NaBH_4$ but not



41. (2)

If X, Y and Z do not form ppt with dil HCl and form ppt with dil H_2S in mineral acid it means cation X, Y and Z belong to 2nd group. X and Y are insoluble in ammonium poly sulphide it means X and Y belongs to 2(1) and Z belongs to 2(2).

42. (4)



and CH_3CHO

43. (4)

Oxymercuration - demarcation gives Markovnikov's addition product but without rearrangement.

44. (3)

None of fac or Mer isomer is optically active, so enantiomers are not possible.

45. (1)

105 - Unp - Unnilpentium

46. (2)

Moles at eq. $\underset{2-x}{P_{2(g)}} + \underset{4-x}{Q_{2(g)}} \rightleftharpoons \underset{2x}{2PQ_{(g)}}$

$K_c = \frac{4x^2}{(2-x)(4-x)} \Rightarrow x = \frac{4}{3}$

$[PQ] = \frac{2 \times 4}{3 \times 4} = \frac{8}{3 \times 4}$ Moles

$= \frac{3 \times 8}{3 \times 4} = 2.0M$

47. (63)

Meq. Of reductant in 25 mL = Meq. Of $KMnO_4$

$= 20 \times 0.01 \times 5$

$\therefore$ Meq. Of reductant in 1 litre

$= 20 \times 0.01 \times 5 \times 40 = 40$

$A_2^{+a} \rightarrow 2A^{+b} + 2e^-$

$\Rightarrow$ n-factor of $A_2^{+a} = 2$

$\therefore$ Meq. of reductant = 40

$\therefore \quad \frac{w}{M/2} \times 1000 = 40$

$\therefore \quad \frac{2.52\times 2\times 1000}{M}=40$

$\therefore \quad M=126; \frac{M}{2}=63$

48. (1)

rate = 10^{-5} [R – X][OH^-] + 10^{-6} [R – X]

% of S_{N_2} reaction

$= \frac{10^{-5}[R-X][OH^-]}{10^{-}[R-X][OH^-]+10^{-6}[R-X]}\times 100$

$= \frac{10^{-5}\times[OH^-]}{10^{-5}[OH^-]+10^{-6}}\times 100$

At [OH^-] = 0.0001

$= \frac{10^{-5}\times 0.001}{10^{-5}\times 0.001+10^{-6}}\times 100$

$= \frac{10^{-8}\times 100}{10^{-8}+100\times 10^{-8}}$

$= \frac{100}{101}=0.99\%$

49. (3)

Moles of $Na_2CO_3.xH_2O$ = $\frac{8}{106+18x}$

If solution is diluted then number of moles of $Na_2CO_3\cdot XH_2O$ remain same.

Moles of $Na_2CO_3\cdot xH_2O$ in 1 L = $\frac{8}{106+18x}$

Moles of $Na_2CO_3.xH_2O$ in

100 mL = $\frac{1}{10}\left(\frac{8}{106+18x}\right)$

2 mol of HCl react with 1 mol of $Na_2CO_3.xH_2O$

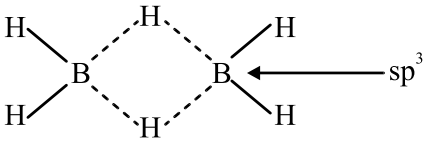
Moles of HCl = $10^{-2}\times 1 = 0.01$

Moles of $Na_2CO_3.xH_2O$

$\Rightarrow 5\times 10^{-3}$ mol

$\frac{8}{1060+180x}=\frac{5}{1000}\Rightarrow x=3$

50. (8)



x = two boron + four hydrogen atom having 2c – 2e bond

y = both boron

SECTION-III (MATHEMATICS)

51. (3)

$f(2x) = f(2x + 1)$

$f(x)=f\left(x+\frac{1}{2}\right)$

So period of $f(x)$ is $\frac{1}{2}$

So $f(4)=f\left(2+\frac{1}{2}\times 4\right)$

$f(6)=f\left(2+\frac{1}{2}\times 8\right)$

$f(8)=f\left(2+\frac{1}{2}\times 12\right)$

So $f(2) = f(4) = f(6) = f(8)$

So value of $f(2) + f(4) – (6) = f(8) = 0$

52. (2)

$f(x)=\sqrt{x-1}+\sqrt{x+24-10\sqrt{x-1}}$

$1\le x\le 26$

$f(x)=\sqrt{x-1}+\sqrt{\left(5-\sqrt{x-1}\right)^2}$

$\therefore \quad \sqrt{x-1}<5$

$f(x) = 5$

53. (4)

For selecting any two numbers out 100 = $^{100}C_2$

$P(S) = {}^{100}C_2$

Now number of ways for getting product of two numbers is divisible by '3' = $^{33}C_2 + {}^{33}C_1\times{}^{67}C_1$

So $P(E)=\frac{{}^{33}C_2+33\times 67}{{}^{100}C_2}=0.55$

(where 33 is total number from 1 to 100 which is multiple of 3)

54. (2)

n(S) = 3 × 3 × 3 × 3 = 81

n(E) = all four elements are 0

+ exactly three elements are 0

+ exactly two elements are 0

+ no element is 0

n(E) = 1 + ($^4C_3 \times {}^2C_1$) + (4 × 2 × 2) + (1 + 1 + 4)

= 1 + 8 + 16 + 6 = 31

55. **(2)**

$x - 8 \geq 0,\ 8 - x \geq 0$

$\Rightarrow x = 8$

when $x = 8$. L.H.S = 0, R.H.S = 0

$\Rightarrow x = 8$ is only solution

56. **(2)**

$$f(x) = e^x, g(x) = x, x > 0$$

$$f(t) = \int_0^t f(t-x) \cdot g(x)\,dx$$

$$= \int_0^t e^{t-x} \cdot x\,dx$$

$$= e^t \int_0^t e^{-x} \cdot x\,dx$$

$$= e^t \left[-xe^{-x}\Big|_0^t - \int_0^t \left(-e^{-x}\right) dx \right]$$

$$= e^t \left(-te^{-t}\right) + e^t \left(-e^{-x}\right)\Big|_0^t$$

$$= -t + e^t \left(-e^{-t} + 1\right)$$

$$= -t - 1 + e^t$$

$$= e^t - (t+1)$$

57. **(2)**

Applying componendo and dividendo we get

$$\frac{dy}{dx} = \frac{e^{-x}}{e^x} = e^{-2x}$$

$$\Rightarrow 2y = -e^{-2x} + C$$

$$\Rightarrow 2ye^{2x} = C\,e^{2x} - 1$$

58. **(2)**

$$1 - \frac{b^2}{a^2} = 1 - \frac{a^2e^2}{b^2} \Rightarrow e^2 \pm e - 1 = 0$$

$$\Rightarrow e = \frac{\sqrt{5}-1}{2}$$

59. **(2)**

$f(g(x)) = x$

$$f'\left(g(x)\right).g'(x) = 1$$

$$g'(2) = \frac{1}{f'(a)}$$

$$= \frac{a^2+1}{a^{10}}$$

60. **(1)**

$$I_n = \int_{-\pi}^{\pi} \frac{\sin nx + 2^x \cos x}{(1+4^x)\sin x}\,dx$$

$$= \int_0^{\pi} \left\{ \frac{\sin nx + 2^x \cos x}{(1+4^x)\sin x} + \frac{-\sin nx + 2^{-x}\cos x}{(1+4^{-x})(-\sin x)} \right\} dx$$

$$= \int_0^{\pi} \frac{\sin nx}{\sin x}\,dx$$

$$\Rightarrow \quad I_{n+2} - I_n = 2\int_0^{\pi} \cos(n+1)x\,dx = 0, \text{if } n \in N,$$

61. **(3)**

$$|z_1 + z_2 + \ldots.z_n| = |(z_1 - 1) + (z_2 - 2) + \ldots + (z_n - n) + (1 + 2 + 3 + \ldots.n)|$$

$$\leq |z_1 - 1| + |z_2 - 2| + \ldots. + |z_n - n| + (1 + 2 + \ldots + n)$$

$$\leq (1 + 2 + 3 + \ldots. + n) + (1 + 2 + 3 + \ldots. + n)$$

$$= 2.\left(\frac{n(n+1)}{2}\right) = n(n+1)$$

$$\Rightarrow |z_1 + z_2 + \ldots.z_n| \leq n(n+1)$$

Hence maximum value of $|z_1 + z_2 + \ldots.z_n|$ is $n(n+1)$.

62. **(3)**

Let 2, 4, 5, 6, 8, 17 is x_i.

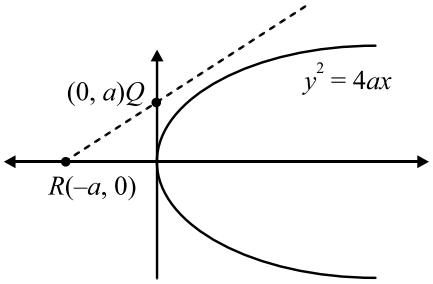
Then 5, 11, 14, 17, 23, 50 can be represented by $(3x_i - 1)$.

$\therefore$ Required variance = 9 × 23.33 = 209.97

63. **(1)**

For $|PR - PQ|$ to be maximum, P should lie on the line segment joining R and Q.

Equation of QR is $\frac{x}{-a} + \frac{y}{a} = 1 \Rightarrow -x + y = a$



$$\Rightarrow (a+x)^2 = 4ax$$

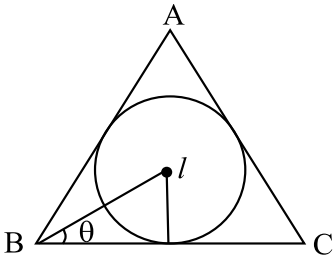
$$\Rightarrow x = a, y = 2a$$

So, line QR, on extending intersects the parabola at M, so M is $(a, 2a)$.

64. **(3)**

Slope of IB, $m_1 = \frac{3-2}{2-1} = 1$

Slope of BC, $m_2 = 2 + \sqrt{3}$



$$\therefore \ \angle IBC = \theta = \tan^{-1}\left|\frac{m_2 - m_1}{1 + m_2 m_1}\right|$$

$$= \tan^{-1}\left|\frac{2+\sqrt{3}-1}{1+\left(2+\sqrt{3}\right)}\right|$$

$$= \tan^{-1}\left|\frac{1+\sqrt{3}}{3+\sqrt{3}}\right|$$

$$= \tan^{-1}\left|\frac{1+\sqrt{3}}{\sqrt{3}\left(1+\sqrt{3}\right)}\right|$$

$$= \tan^{-1}\left|\frac{1}{\sqrt{3}}\right|$$

$= 30°$

$\therefore \angle ABC = 2\theta = 60^o$

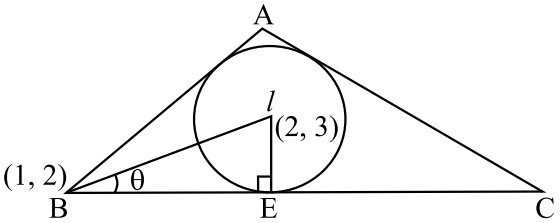
Second method :

$$\sin\theta = \frac{IE}{BI}$$

$$IE = \frac{\sqrt{3}+1}{2\times\sqrt{\sqrt{3}+2}}$$

$$BI = \sqrt{2}$$

$$\sin\theta = \frac{\sqrt{3}+1}{2\sqrt{2}\times\sqrt{\sqrt{3}+2}} = \frac{\sqrt{3}+1}{2\sqrt{4+2\sqrt{3}}} = \frac{1}{2}$$



$y = \left(2+\sqrt{3}\right)x - \sqrt{3}$

$\theta = 30°, 2\theta = 60°$

65. (4)

$$x^{15} + \frac{15}{x} = x^{15} + \left(\frac{1}{x} + \frac{1}{x} + ...15 times\right)$$

Applying A.M – G.M inequality

$$\frac{x^{15} + \left(\frac{1}{x} + \frac{1}{x} + ... + 15\ times\right)}{16} \geq \left(x^{15}\cdot\frac{1}{x}\cdot\frac{1}{x}.....\frac{1}{x}\right)^{\frac{1}{16}}$$

$$\Rightarrow \frac{x^{15} + \frac{15}{x}}{16} \geq 1$$

Minimum value of $x^{15} + \frac{15}{x}$ is 16.

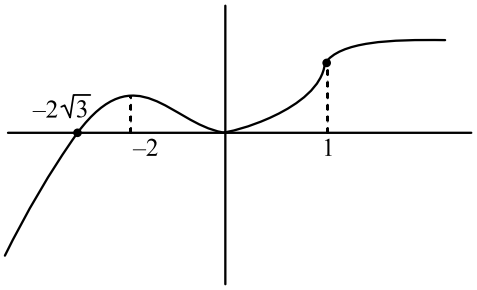
66. (2)

$$(1+20)^{100} = 1 + {}^{100}C_1(20)^1 + {}^{100}C_2(20)^2 + ...$$

$\Rightarrow 01$

67. (3)

$y = f(x)$



68. (1)

$\because$ (1, 2) $\in R$ and (2, 1) $\notin R$

$\Rightarrow R$ is reflexive but not symmetric

69. (4)

$$|\vec{a}+\vec{b}+\vec{c}| = \sqrt{9+15+25}$$

$$\Rightarrow |\vec{a}+\vec{b}+\vec{c}| = 5\sqrt{2}$$

70. (1)

$$I = I_1 - 2I_2 = \int\frac{dx}{e^x + \frac{8}{e^x} + \frac{4}{e^{3x}}} - 2\int\frac{dx}{e^{3x} + 8e^x + \frac{4}{e^x}}$$

$$= \int\frac{(e^{3x} - 2e^x)dx}{e^{4x} + 8e^{2x} + 4}$$

$$= \int\frac{e^x(e^{2x} - 2)dx}{(e^x)^4 + 8(e^x)^2 + 4}, \text{ put } e^x = t$$

$$\Rightarrow \quad e^x\,dx = dt$$

$$= \int\frac{(t^2 - 2)dt}{t^4 + 8t^2 + 4}$$

$$= \int\frac{\left(1 - \frac{2}{t^2}\right)dt}{t^2 + 8 + \frac{4}{t^2}} = \int\frac{\left(1 - \frac{2}{t^2}\right)dt}{\left(t + \frac{2}{t}\right)^2 - 2t\cdot\frac{2}{t} + 8}$$

$$= \int\frac{\left(1 - \frac{2}{t^2}\right)dt}{\left(t + \frac{2}{t}\right)^2 + 4} \text{ put } t + \frac{2}{t} = z$$

$$\Rightarrow \quad \left(1 - \frac{2}{t^2}\right)dt = dz$$

$$= \int\frac{dz}{z^2 + 4} = \frac{1}{2}\tan^{-1}\left(\frac{z}{2}\right) + c$$

$$= \frac{1}{2}\tan^{-1}\left(\frac{t + \frac{2}{t}}{2}\right) + c$$

$$= \frac{1}{2}\tan^{-1}\left(\frac{t^2 + 2}{2t}\right) + c = \frac{1}{2}\tan^{-1}\left(\frac{e^{2x} + 2}{2e^x}\right) + c$$

71. **(15)**

Six number in A.P. are $a - 5d$, $a - 3d$, $a - d$, $a + d$, $a + 3d$, $a + 5d$

according to question $6a = 3$

$$a = \frac{1}{2}$$

Also, $(a - 5d) = 4(a - d)$

$a - 5d = 4a - 4d$

$3a = -d$

$$d = -\frac{3}{2}$$

Terms of A.P. are 8, 5, 2, –1, –4, –7

72. **(7)**

$$\frac{x^3}{(x-z)(x-y)} + \frac{y^3}{(y-x)(y-z)} + \frac{z^3}{(z-x)(z-y)}$$

$$= \frac{(x+y+z)(x-y)(y-z)(z-x)}{(x-y)(y-z)(z-x)} = x + y + z$$

$\therefore$ Total number of possible values = 7

73. **(5)**

$Y = x^3 - 2x^2 + x + 5$

$\Rightarrow f(0) = 5$

74. **(10)**

$\overrightarrow{OP} \cdot \hat{i} = 2 \quad \Rightarrow \quad x = 2$

$P \equiv (2, 125)$ & $Q \equiv (-2, -115)$

75. **(0)**

$$\frac{dy}{dx} - \frac{y}{x^2}\tan\left(\frac{1}{x}\right) = -\sec\left(\frac{1}{x}\right) \times \frac{1}{x^2} \text{(linear)}$$

$$\therefore \; e^{\int -\frac{1}{x^2}\tan\frac{1}{x}dx} = \sec\left(\frac{1}{x}\right)$$

$$\therefore \text{ solution } y\sec\frac{1}{x} = \tan\frac{1}{x} + k$$

$\because \; y \to -1$ as $x \to \infty$

$\therefore \; k = -1$

$$\Rightarrow \; y = \sin\frac{1}{x} - \cos\frac{1}{x}$$



PW Web/App - https://smart.link/7wwosivoicgd4

Library- https://smart.link/sdfez8ejd80if